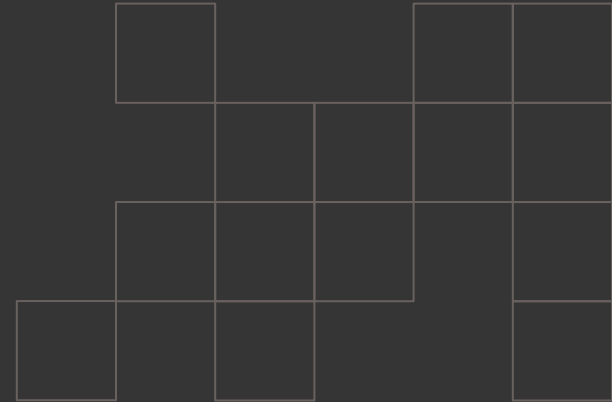


Ambrosia Ardenvale-Wood
2026 / Winter / BUS 267

EIA Renewable Energy Analysis

U.S. Solar, Wind & Hydroelectric Power Time Series Model



Executive Summary

The Bottom Line



50+ YEARS

Historical Data

$R^2 = 0.88$

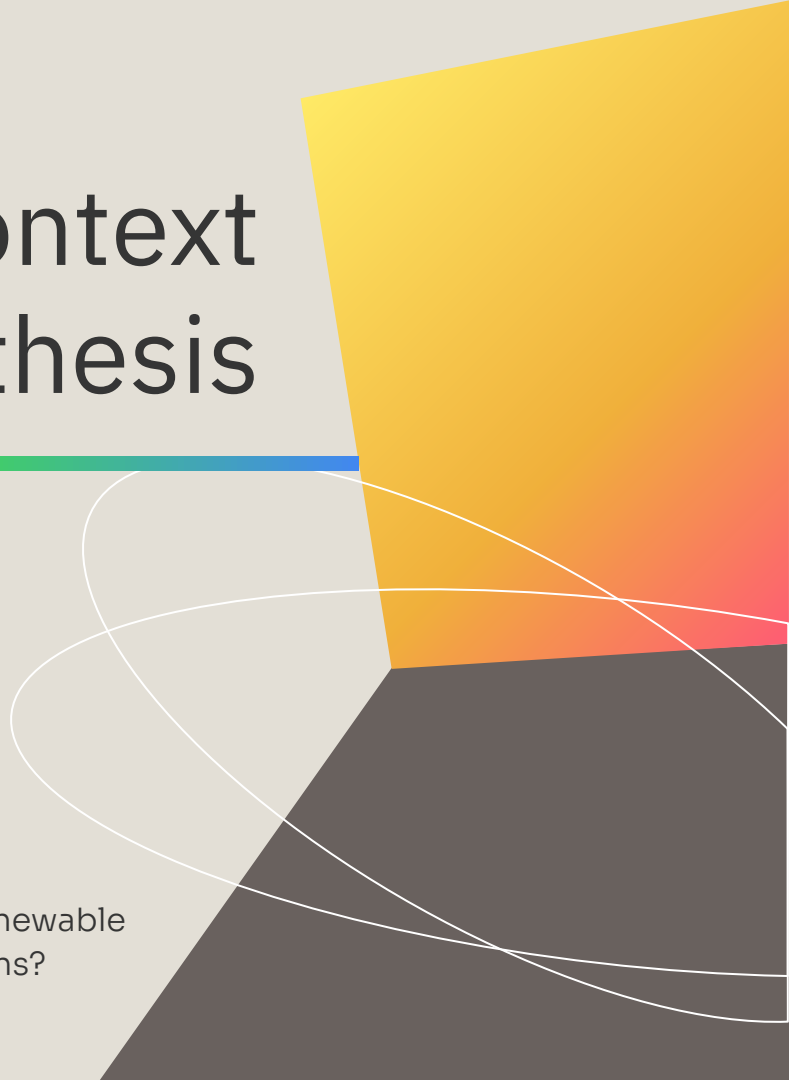
88% Model
Accuracy

Spring & Summer

Primary Growth Drivers



Business Context & Hypothesis



Stakeholder

U.S. Energy Investment Firm

Objective

Optimize renewable energy portfolio planning

Hypothesis

Can seasonal cycles and annual growth trends predict total renewable consumption to support grid stability and investment decisions?

Dataset Overview

Source

EIA Monthly Energy Review, Table 10.1

Dataset Details

634 monthly observations used across
Solar, Wind & Hydro (1973 - 2025)

Key Cleaning Steps

- Filtered to (3) renewable sources
- Removed annual summary rows
- Parsed monthly dates
- No duplicates found

Solar & Wind

Early-year zeros reflect absence of
utility infrastructure during timeframe

Feature Engineering

Season, Year, Total Renewables,
polynomial & interaction terms

Renewable Energy Trends

Hydro

Stable & Declining → Climate & Drought.

Wind

Rapid Growth. Dominant Renewable Source.

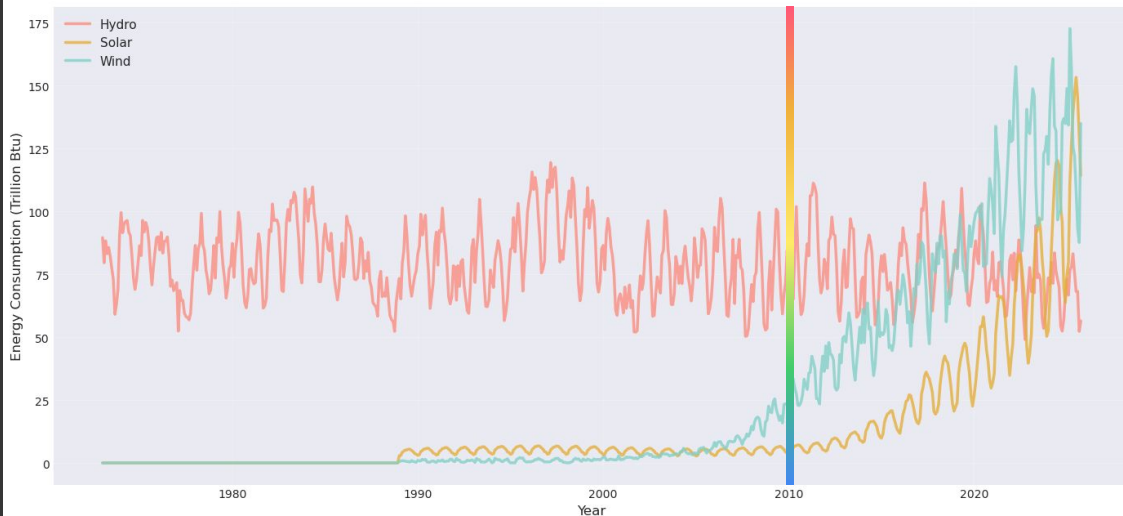
Solar

Exponential Growth. Steepest Trajectory.

2010

Inflection Point → Technology Costs Plummeted

U.S. Renewable Energy Consumption: Solar, Wind, and Hydro (1973-2025)



Seasonal Patterns

Key Findings

Hydro Peaks Spring →
Snowmelt Driven

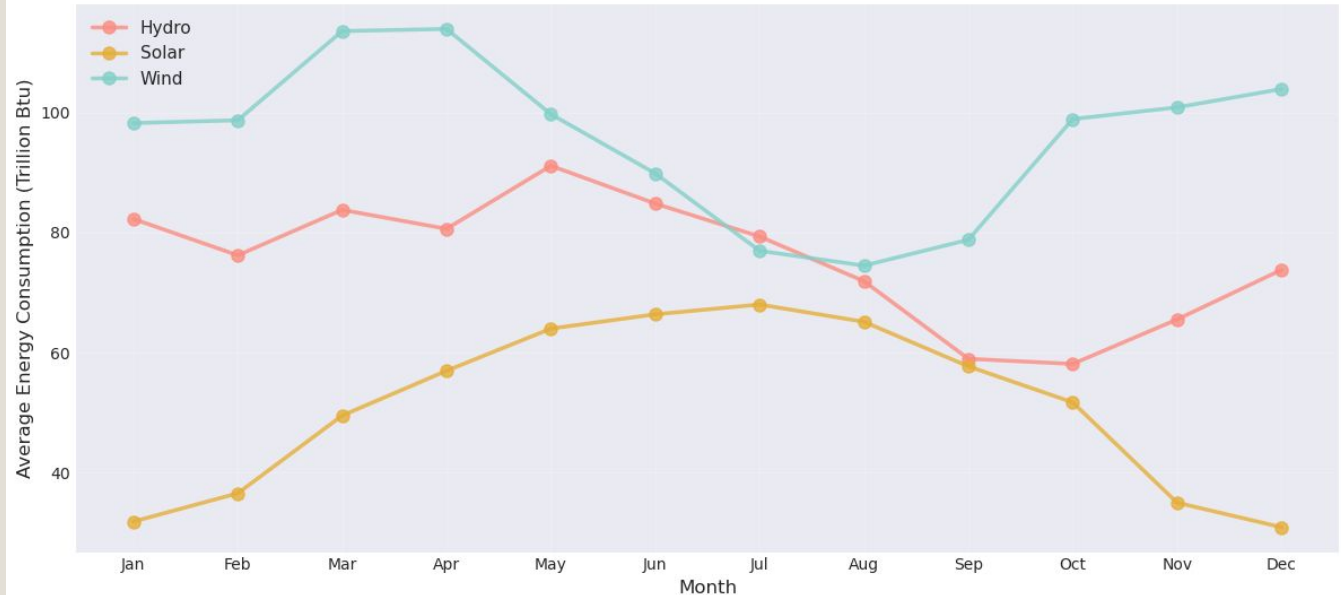
Solar Peaks Summer →
Maximum Daylight Hours

Wind Peaks Winter →
Atmospheric Conditions

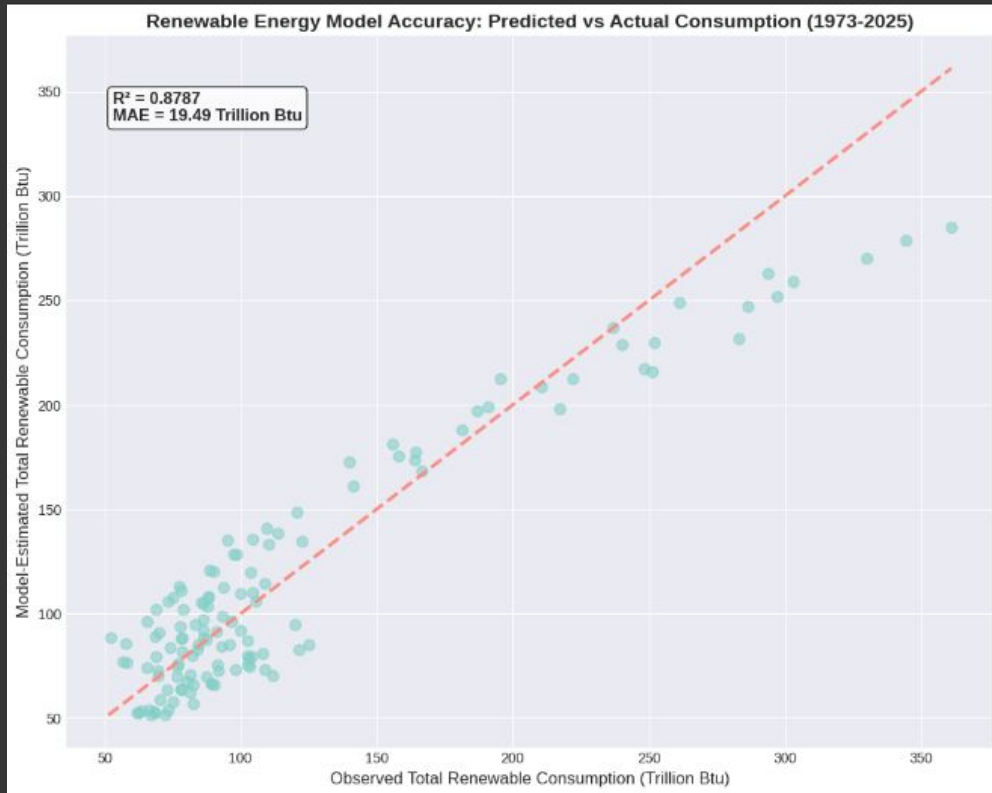
Natural Synergy → Provides
Year-round Coverage

Intermittency Gap → Sept & Oct
sit between Summer solar &
Winter wind peaks

Average Monthly Renewable Energy Consumption (2015-2025)



Regression Model & Results



Model

Linear Regression
with polynomial &
interaction terms

Dependent Variable

Total Renewable
Consumption
(Trillion Btu)

Independent Variables

Year, Season & Month

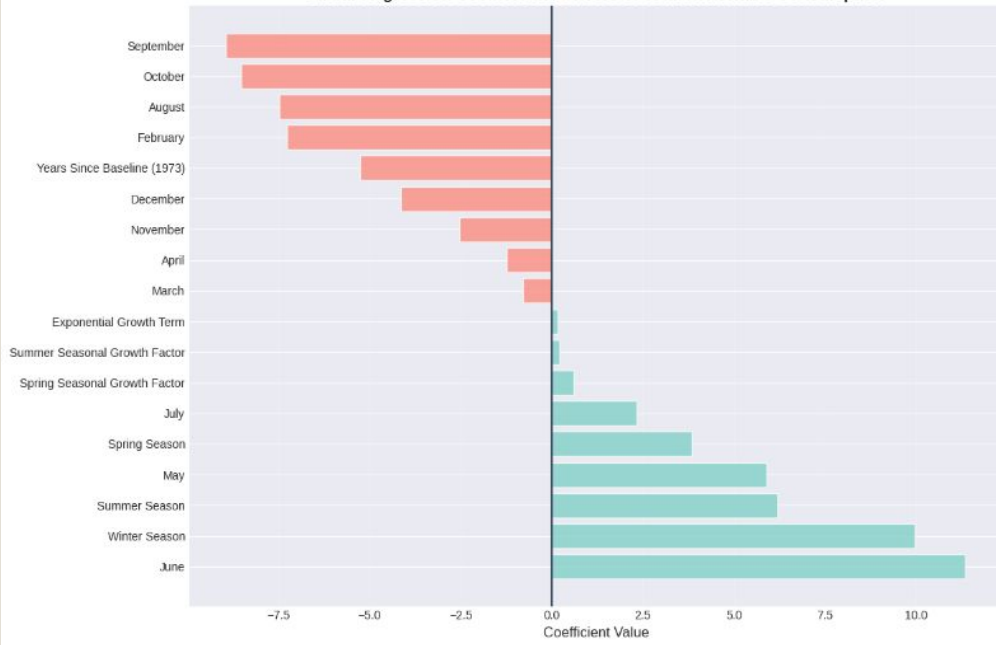
R-Squared

0.88 → Explains 88% of
Variance

MAE

19.49 Trillion Btu

Linear Regression Coefficients: Drivers of Total Renewable Consumption



Key Findings

June single strongest month driver (+11.35)

Winter & Summer seasons both elevate total consumption

Spring Seasonal Growth Factor (+0.61)
seasonal peaks intensifying each year

Sept & Oct → consistent annual trough
(-8.94 & -8.53)

Coefficient Insights

Business Recommendations

Strategic Capacity Planning

Schedule maintenance in late Autumn to maximize asset availability during Spring & Summer high performance window

Storage Investment

Featured

Sept through Oct is a storage problem. Recommend increasing storage to capture July energy production. Target battery & pumped hydro solutions.

Model Implementation

R^2 of 0.88 makes this model viable as a secondary validation tool for 12-month budgetary forecasting.

U.S Renewable Energy Landscape

Conclusion

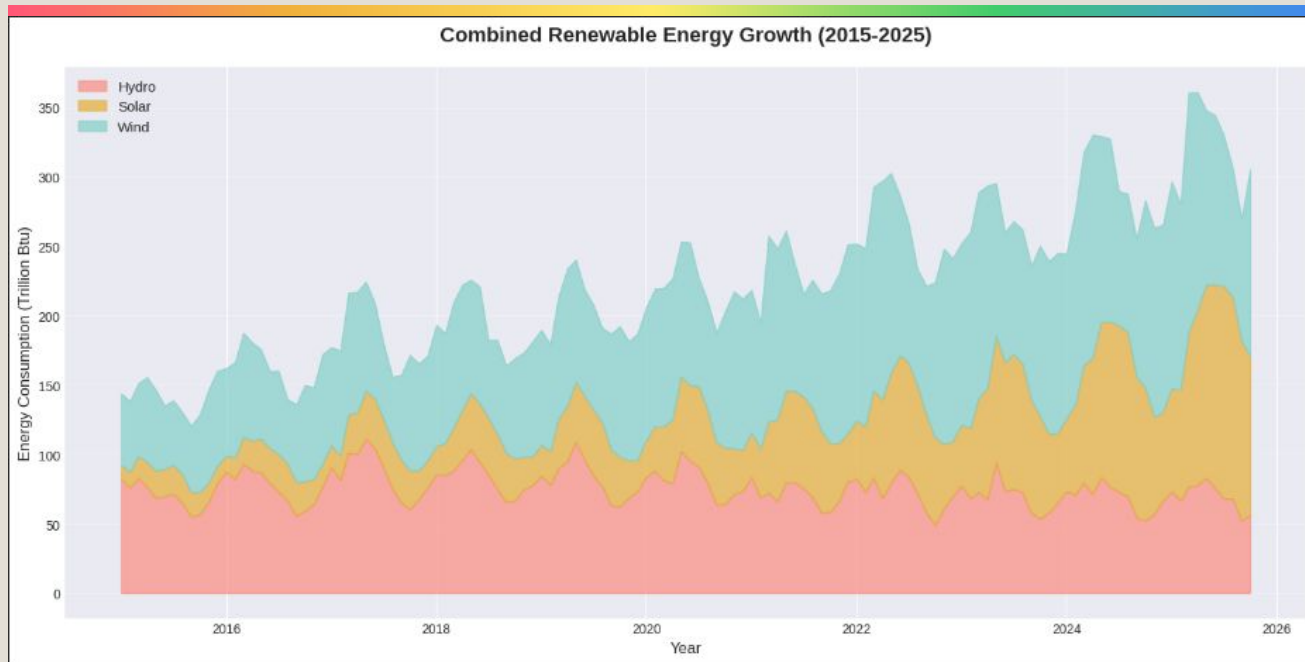
50+ years of EIA data analyzed across solar, wind & hydro.

Exponential growth in solar & wind confirmed analytically.

Seasonal balance validated. Diversified portfolio more stable.

Regression model achieves $R^2 = 0.88$ using temporal features.

Solar: Most growth potential.
Wind: Most predictable.
Hydro: Stable baseline.





Thank you